PRISM: A Multimodal Pipeline for Post-Treatment Glioblastoma Surveillance with Interpretable Clinical Reporting

Sri D. Bandreddi, Amisha Dhawan, Yelim Kim, Freya Wang and Ethan Xin

***Abstract*—Post-treatment surveillance of glioblastoma (GBM) remains one of the most clinically difficult problems in neuro-oncology. After chemoradiation, MRI changes from true tumor progression and treatment-related effects such as pseudoprogression are nearly impossible to distinguish by eye, yet the consequences of misclassification are severe. Current standard-of-care criteria (RANO) rely on visual comparison of contrast-enhancing lesions and cannot reliably resolve this ambiguity. Machine learning models have also attempted to resolve this problem, yet no existing system integrates multimodal progression prediction with structured, interpretable physician-facing output. We present PRISM ((Post-treatment Response Intelligence and Surveillance Monitor), an end-to-end pipeline for post-treatment GBM surveillance. A Swin UNETR imaging encoder, a calibrated radiomics model, and a Late Fusion MLP predict tumor progression; then a three-layer agentic system synthesizes model outputs with RANO 2.0 criteria to generate clinical reports. The fusion model achieved macro AUC-ROC of 0.80 and overall accuracy of 80.95%, outperforming the molecular-only baseline (macro F1 0.74). Discordance between RANO and model outputs is surfaced explicitly to give clinicians actionable context rather than a black-box prediction. PRISM represents a step toward interpretable, clinician-facing AI for post-treatment GBM monitoring.**

***Clinical Relevance*— The current standard RANO criteria cannot distinguish true progression from treatment-related changes. PRISM is able to provide clinicians with structured, interpretable progression assessments, directly supporting more informed treatment decisions.**

I. Introduction

Glioblastoma (GBM) is the most aggressive primary malignant brain tumor in adults, with an incidence of 3.2–5 per 100,000 people and a 5-year survival rate of approximately 5% [1][2]. Despite surgical resection followed by concurrent chemoradiation with temozolomide [4], median overall survival remains approximately 15 months [3]. After treatment, clinicians face a high-stakes diagnostic challenge: whether imaging changes represent true tumor progression or treatment-related effects such as radiation necrosis and pseudoprogression. GBM's intrinsic heterogeneity, combined with the tissue changes caused by chemoradiation, produces overlapping MRI patterns that look nearly identical whether the tumor is growing or dying [2][5].

The current standard for post-treatment response assessment is the Response Assessment in Neuro-Oncology (RANO) criteria, which evaluates progression through visual comparison of contrast-enhancing lesions across T1, T1c, T2, and FLAIR MRI sequences over time. T1 post-contrast highlights actively enhancing tumor tissue; T2 captures overall lesion; and FLAIR represents infiltrating non-enhancing tumor and peritumoral edema [6]. Despite widespread adoption, RANO lacks the ability to capture pseudoprogression, which affects 9–36% of post-treatment GBM patients [7][8] and has a pooled incidence of 36% in newly diagnosed GBM (95% CI 33–40) [9]. Misclassification leads to unnecessary treatment changes, premature discontinuation of effective therapies, and immense psychological burden [10].

Machine learning and deep learning have been widely applied to GBM prognosis using clinical, molecular, imaging, and radiomics data [1], with Patricoski-Chavez et al. achieving an AUROC of 0.82 for early glioma recurrence prediction [11]. However, none of these models have been deployed in clinical care. The MU-Glioma-Post dataset [5], introduced by Mahmoud et al. specifically for post-treatment AI model development, has been used by three recent studies: Lteif et al. proposed a new segmentation augmentation strategy [12]; Saridede et al. developed a GBM vs non-GBM classification model [13]; and Christodoulou et al. built a calibrated radiomics model for progression prediction [2]. While Christodoulou et al.'s model produces a probability for progression, it is without a structured clinical context. Therefore, there has yet to be an LLM-based clinical agent that delivers a structured, interpretable report to the treating physician.

We propose PRISM, an end-to-end pipeline with a model for probabilistic progression prediction paired with an AI agent that synthesizes model outputs and clinical features into a structured physician-facing report. PRISM integrates four components: a custom Swin UNETR (Swin UNEt Transformer) imaging encoder; a calibrated logistic regressions radiomics model; a Late Fusion MLP combining imaging and molecular evidence; and a three-layer agentic report generation system. On a patient-aware held-out test set, the fusion model achieved a macro AUC-ROC of 0.80, macro F1 of 0.77, and overall accuracy of 80.95%, demonstrating reliable discrimination between true progression and treatment-related effects. The agentic report produced all four possible concordant outcomes accordingly and handled RANO-model disagreement with clinical context, showing feasibility for clinical deployment.

II. Methods

A. Dataset

We used MU-Glioma-Post [5], a publicly available longitudinal post-operative glioma cohort on The Cancer Imaging Archive comprising 203 patients and 594 post-treatment MRI timepoints. The dataset includes T1, T1 post-contrast, T2, and FLAIR sequences with standardized preprocessing and automated tumor segmentations. GBM accounts for 77.34% of cases. Patients with progression represent approximately 75% of the cohort, reflecting a 3:1 class imbalance of progressors to non-progressors. Timepoints per patient range from 1 to 6, with a mean of about 2.9. Model evaluation used a patient-aware train/validation/test split ensuring no timepoints from the same patient appeared across partitions.

B. MRI Feature Extraction via Swin UNETR

MRI feature extraction was performed using a Swin UNETR-based imaging encoder. Swin UNETR was selected because it has shown strong performance in brain tumor segmentation tasks and can model both local tumor structure and broader 3D spatial context through shifted-window attention [14]. Given the limited number of scans and the post-operative nature of the dataset, a BraTS-pretrained Swin UNETR model was used as the starting point and fine-tuned on the study cohort to better adapt the learned tumor representations to this setting.

Each MRI scan was preprocessed as follows. The four modality volumes (T1, T1-CE, T2, FLAIR) were loaded using nibabel and concatenated into a 4-channel 3D array. Voxel intensities were clipped to the 1st–99th percentile and rescaled to [0, 1] per modality to remove scanner artifacts while preserving tissue contrast. The segmentation mask was used to crop the tumor bounding box, expanded by 10 voxels in all directions to include peritumoral tissue. The cropped volume was resampled to $128 \times 128 \times 128$ voxels as required by the Swin UNETR architecture.

We implemented a custom Swin UNETR encoder consisting of three components. First, a 3D convolutional patch embedding layer partitions the volume into non-overlapping $4 \times 4 \times 4$ patches, projecting each into a 48-dimensional vector and yielding 32,768 patch tokens. Second, a 4-layer transformer with multi-head self attention processes all tokens embeddings together, allowing the model to relate distant regions of the tumor volume to one another regardless of their spatial separation. Third, global average pooling compresses all 32,768 patch embeddings into a single 48 dimensional imaging embedding, which is passed to the fusion layer. Encoder outputs were reviewed using coverage heatmaps, PCA visualization of the 594 scan-level embeddings, and temporal trajectory analysis to assess whether the learned representations captured meaningful spatial and longitudinal variation.

C. Radiomics-Based Progression Prediction

We used Optuna to search across candidate model families including LightGBM, logistic regression, random forest, and SVM across multiple feature subsets. The strongest exported model was a calibrated logistic regression and was designed to predict whether tumor progression would occur within 120 days of a given post-treatment scan, using 48 selected features: 23 T1c, 19 FLAIR, and 6 clinical.

Prior to extraction, N4 bias field correction was applied to remove scanner-induced intensity gradients. The lesion ROI was defined precisely from segmentation labels 1, 2, and 3, with the resection cavity explicitly excluded to prevent post-surgical artifacts from distorting measurements. Feature extraction focused on T1-CE for its sensitivity to actively enhancing tumor tissue and FLAIR for non-enhancing spread and peritumoral edema. For each sequence, features were extracted from the original image as well as LoG and Wavelet filtered versions to capture tumor boundary characteristics and multi-scale texture patterns.

Feature selection was performed on training data only, comprising variance filtering, Spearman correlation filtering, and ranking by L1 logistic regression coefficients under patient-aware cross-validation. The final feature set included 23 T1-CE radiomic features, 19 FLAIR radiomic features, and 6 clinical features, including age, sex, and molecular markers such as IDH1, MGMT methylation, and TERT status, selected for their established prognostic relevance in GBM [3].

Finally, Platt calibration was applied post-training so that output scores reflect true progression likelihoods suitable for threshold based decision making. Evaluation used patient aware 5-fold cross-validation to prevent timepoint level leakage across longitudinal scans.

D. Multimodal Late Fusion MLP

To combine imaging and radiomic features into a single progression prediction, we implemented a Late Fusion MLP. Late fusion was chosen over early fusion to preserve each modality's distinct feature characteristics. Merging imaging embeddings and radiomic features at the input level would force the network to learn from two very different feature spaces simultaneously, likely degrading performance. Therefore, each modality was processed through a separate encoding branch before concatenation. Following the ModDrop strategy of Neverova et. al [15], a 30% modality dropout layer was applied during training to maintain robustness when one modality is unavailable, reflecting the clinical reality of incomplete data at some patient timepoints. Focal Loss was used to address the 3:1 class imbalance by concentrating learning from the minority progression cases. The classification threshold was set at 0.48 to favor recall over precision as missing a true progression presents greater clinical consequences than a false alarm.

E. Clinical Report Generation via Agentic Pipeline

The report generation system is a three-layer pipeline that synthesizes all model outputs into a structured clinical report

for physicians to review.

Layer 1 implements RANO 2.0 criteria as hard-coded rules, evaluating three independent criteria per timepoint: Criterion A (bidimensional product $\geq$25% above nadir), Criterion B (enhancing tumor volume relative to ET nadir), and Criterion C (SNFH, or non-enhancing FLAIR signal, volume $\geq$25% above nadir). Nadir values are computed exclusively from prior timepoints to prevent the model using future information; at a patient's first visit, RANO assessment is marked not applicable. Each criterion is independently reported, as an isolated SNFH trigger carries different clinical meaning than a multi-criterion positive.

Layer 2 reconciles RANO and ML outputs through a four-state concordance check: RANO+/AI+, RANO−/AI−, RANO+/AI−, and RANO−/AI+. Discordant cases are flagged for human review rather than automatically resolved. Concurrently, a pseudoprogression confounder score is computed from time since radiotherapy, MGMT methylation status, IDH1 status, steroid use, and scan timing, setting a pseudo_suspected flag that governs which report template Layer 3 uses.

Layer 3 passes a fully deterministic structured prompt to GPT-4.1-mini, which was selected for instruction-following reliability and low hallucination rate in structured generation, to produce the clinical report. The model receives only pre-computed values from Layers 1 and 2 and cannot introduce information beyond what was already validated. Four prompt branches cover concordant progression, concordant stability, discordant with low pseudoprogression risk, and discordant with elevated pseudoprogression suspicion. A consistency check verifies seven properties before every report is delivered: nadir integrity, threshold scope, criterion logic, concordance consistency, modality labeling, percentage correctness, and SHAP attribution.

III. Results

A. Encoder Validation

Coverage verification confirmed successful embedding generation across all 594 timepoints. PCA of the resulting 594 embeddings showed that PC1 and PC2 captured 80.1% of total variance (66.1% and 14.0% respectively), indicating that the encoder produced structured, informative representations rather than arbitrary outputs. Temporal trajectory analysis of the five patients with the most timepoints showed diverging PC1 trajectories across visits, suggesting the encoder captures meaningful longitudinal change rather than producing static per-patient representations.

B. Radiomics Model

The calibrated logistic regression model achieved an AUC-ROC of 0.8043, accuracy of 0.8095, balanced accuracy of 0.7243, and macro F1 of 0.7375. At the operating threshold of 0.4553, the model achieved a positive-class F1 of 0.6000, precision of 0.6667, and recall of 0.5455. The Brier score of 0.1542 confirms well-calibrated probability outputs, meaning the model's scores reflect true progression likelihoods rather than arbitrary confidence values. Feature importance was imaging-led (T1c 55.9%, FLAIR 31.1%, clinical 13.0%) consistent with prior literature on the importance of multiparametric MRI in post-treatment GBM assessment [2].

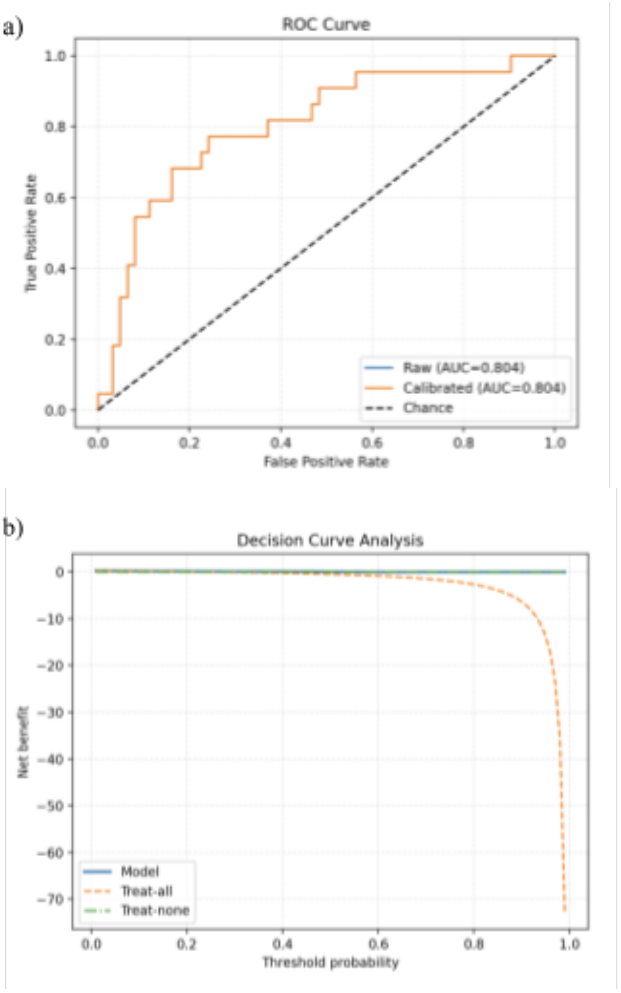


Figure 1. Held-out ROC (a) and decision-curve analysis (b) for the exported corrected forward model. The model retained good ranking performance over a good threshold range.



Figure 2. Peak-model SHAP shared by input block. The lean hybrid model is dominated by T1c, complemented by FLAIR, and only lightly supported by clinical covariates.

C. Fusion Model

The Late Fusion MLP achieved overall accuracy 80.95%, AUC 0.801, and macro F1 0.7683, representing a meaningful improvement over the molecular-only baseline of 0.7375. At the 0.48 threshold, recall was 76.05%, confirming that

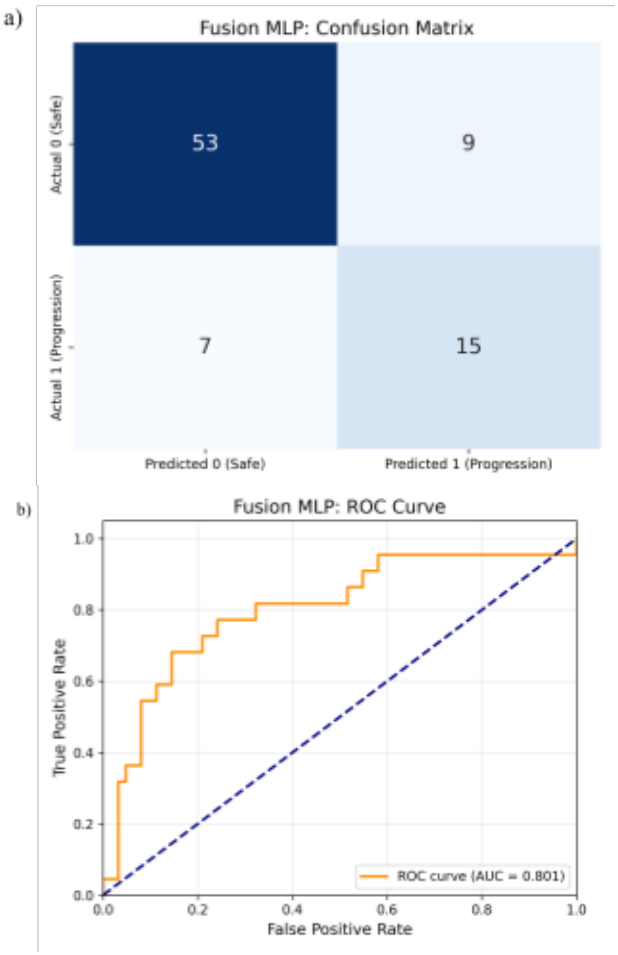


Figure 3. Confusion matrix (a) and ROC curve (b) of the Late Fusion MLP, demonstrating improved recall over the molecular-only baseline at a 0.48 classification threshold.

imaging features enabled the model to identify progression cases missed by the molecular baseline alone.

D. Agentic Report System

The pipeline successfully handled all four concordance outcome types across all cases. Six of seven automated consistency checks returned PASS on an illustrative sample report seen in figure 4; one was marked NOTED for global rather than per-prediction SHAP attribution, a known limitation currently under active development. In this illustrative RANO+/AI− case (PatientID_0053, Timepoint 6), SNFH increased 38.4% from nadir at day 521 post-treatment, triggering Criterion C, while the Fusion MLP returned progression probability 0.3047. The pseudoprogression confounder score was LOW given the elapsed time since radiotherapy, and the report framed the clinical question as whether the T2/FLAIR increase represented genuine infiltrative spread not captured by the fusion model.

IV. Discussion

The macro F1 improvement from 0.7375 (molecular only) to 0.7683 (fusion) demonstrates that imaging and molecular data provide complementary rather than redundant information for post-treatment progression monitoring. Notably, the largest performance gains in the radiomics branch came not from increasing model complexity but from redefining the prediction task. Shifting from classifying whether a scan was already at progression to predicting whether progression would occur within 120 days produced a jump in AUC from 0.599 to 0.804, underscoring that careful problem formulation is as important as model selection in clinical machine learning. At the fusion level, the imaging-led feature importance profile aligns with prior literature emphasizing contrast-enhancing tissue and non-enhancing tumor spread as the primary indicators of post-treatment GBM status. Additionally, the fusion model's ability to correct incorrect molecular predictions in edge cases further confirms imaging as an independent validation signal.

Beyond predictive performance, the pipeline treats RANO-vs-AI disagreement as clinically meaningful rather than resolving it silently. When rule-based and model-based assessments disagree, flagging this disagreement alongside pseudoprogression risk gives clinicians more actionable information than either assessment alone. This is particularly important given that aforementioned pseudoprogression affects up to 36% of newly diagnosed GBM patients and that misclassification in either direction carries significant clinical consequences [9][10].

Finally, the separation of RANO rule evaluation, probabilistic inference, and report generation into distinct layers enables the transparency that clinical AI tools require. Each layer exposes its intermediate outputs, allowing clinicians to verify RANO calculations, inspect ML confidence, and confirm concordance logic before acting on the report. This design directly addresses the FDA's 2024 Guiding Principles for Transparency in Machine Learning-Enabled Medical Devices [16].

However, before clinical deployment can be considered, several limitations must be addressed. First, the Swin UNETR encoder was initialized with random weights rather than pretrained BraTS21 weights and was not fine-tuned on MU-Glioma-Post. As a result, domain shift between pre-operative training data and post-treatment MRI, particularly the presence of resection cavities, may limit current feature quality. Loading pretrained weights is the next step. Additionally, global average pooling discards spatial location information that may be clinically relevant for GBM, and future work will explore attention-based pooling strategies. Second, radiomics features are fundamentally intensity- and texture-centric representations, making them useful for capturing lesion heterogeneity but potentially missing explicit anatomical or volumetric structure that segmentation-derived features may capture more directly. More broadly, the fusion model and classification threshold

Glioma Surveillance Report Patient: PatientID_0053			
DISCORDANT — RANO+ / AI− SNFH Criterion C met ∣ Fusion MLP below threshold ∣ Adjudication required		Fusion MLP prob **0.3047** Threshold 0.48 — NOT met	
KEY FINDINGS			
SNFH / T2-FLAIR 14.12 cc vs nadir: **+38.4%** → RANO+ Criterion C	Enhancing Tumor (ET) 3.87 cc vs nadir: +60.8% ∣ BD: 1.35 Criterion B: not flagged	RANO 2.0 Verdict POSITIVE Criterion C only (SNFH) Single-criterion trigger	Pseudo Risk LOW 0.5 Day 521 — pseudo window long closed
CLINICAL CONTEXT			
Age / Sex	68 y / Female	Diagnosis	Glioblastoma, IDH-wildtype
Grade	Grade 4 (WHO 2021)	Surgery	Subtotal resection
MGMT	Methylated	IDH1 / ATRX	Wildtype / Intact
RT Window	Day 19–61 (completed)	Steroids	None (discontinued day 95)
AI INFERENCE & CONCORDANCE			
Radiomics Score (MLP input)		Fusion MLP Output	Concordance
0.1978 Calibrated radiomics probability *MLP input only — 0.48 threshold NOT applicable*		0.3047 Threshold 0.48 → NO ALERT *Real value from test-set inference*	RANO+ / AI− Discordant *Multidisciplinary adjudication required*
AUTOMATED CONSISTENCY CHECKS			
Check		Status	
Nadir window integrity		PASS	
Threshold scope		PASS	
Criterion firing logic		PASS	
Concordance consistency		PASS	
Modality labeling		PASS	
RANO nadir value correctness		PASS	
SHAP source attribution		**NOTED**	

Table 1. Selected sections of an illustrative PRISM report for PatientID_0053, Timepoint 6 (RANO+/AI- discordant case)

have not been validated on external cohorts, limiting generalizability. The class imbalance and predominantly white patient cohort (94.1%) further restrict how broadly these results can be applied. Per-prediction SHAP attribution is planned to replace current global averages, and the generated report has not been systematically evaluated against clinician-written reports. Ultimately, external multi-center validation and a prospective pilot study with radiologist and oncologist feedback are required before this system can be considered for clinical use.

V. Conclusion

Distinguishing true GBM progression from treatment-related imaging changes remains one of the most challenging and consequential problems in neuro-oncology. PRISM addresses this by integrating MRI-derived spatial features, radiomic texture, and molecular data within a pipeline that produces structured physician-facing reports. The fusion model's improvement over the molecular-only baseline confirms that these modalities provide complementary rather than redundant information. Additionally, the three-layer architecture demonstrates that clinical AI tools can be both accurate and transparent. GBM's lethality demands better tools, and PRISM provides a meaningful step toward multimodal, interpretable post-treatment surveillance.

Acknowledgment

The authors thank Professor Patjanaporn Chalacheva and teaching assistant Sid Roy for their guidance and support throughout this project. The authors also thank the University of Missouri and the Cancer Imaging Archive for making the MU-Glioma-Post dataset publicly available.

Author Contributions

A.D. designed and implemented the Swin UNETR imaging encoder. S.D.B. designed and implemented the radiomics pipeline. F.W. designed and implemented the Late Fusion MLP. E.X. designed and implemented the agentic report generation system. Y.K. investigated the clinical relevance of features, and wrote the manuscript with input from all authors. All authors contributed equally.

REFERENCES

[1] R. Poursaeed, M. Mohammadzadeh, and A. A. Safaei, "Survival prediction of glioblastoma patients using machine learning and deep learning: a systematic review," BMC Cancer, vol. 24, no. 1, p. 1581, Dec. 2024, doi: 10.1186/s12885-024-13320-4.

[2] R. C. Christodoulou, G. Vamvouras, R. Pitsillos, E. E. Solomou, and M. F. Georgiou, "Explainable radiomics with probability calibration for postoperative glioblastoma surveillance," Eur. J. Radiol. Artif. Intell., vol. 5, p. 100074, Mar. 2026, doi: 10.1016/j.ejrai.2026.100074.

[3] D. Sipos et al., "Glioblastoma: Clinical Presentation, Multidisciplinary Management, and Long-Term Outcomes," Cancers, vol. 17, no. 1, p. 146, Jan. 2025, doi: 10.3390/cancers17010146.

[4] R. Stupp et al., "Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma," N. Engl. J. Med., vol. 352, no. 10, pp. 987–996, Mar. 2005, doi: 10.1056/NEJMoa043330.

[5] E. Mahmoud et al., "MU-Glioma Post: A comprehensive dataset of automated MR multi-sequence segmentation and clinical features," Sci. Data, vol. 12, no. 1, p. 1847, Nov. 2025, doi: 10.1038/s41597-025-06011-7.

[6] P. Y. Wen et al., "Updated Response Assessment Criteria for High-Grade Gliomas: Response Assessment in Neuro-Oncology Working Group," J. Clin. Oncol., vol. 28, no. 11, pp. 1963–1972, Apr. 2010, doi: 10.1200/JCO.2009.26.3541.

[7] M. J. Belue et al., "Diagnosing Progression in Glioblastoma—Tackling a Neuro-Oncology Problem Using Artificial-Intelligence-Derived Volumetric Change over Time on Magnetic Resonance Imaging to Examine Progression-Free Survival in Glioblastoma," Diagnostics, vol. 14, no. 13, p. 1374, Jun. 2024, doi: 10.3390/diagnostics14131374.

[8] S. C. Thust, M. J. van den Bent, and M. Smits, "Pseudoprogression of brain tumors," J. Magn. Reson. Imaging, vol. 48, no. 3, pp. 571–589, Sep. 2018, doi: 10.1002/jmri.26171.

[9] "Differentiation Between True Tumor Progression of Glioblastoma and Pseudoprogression Using Diffusion-Weighted Imaging and Perfusion-Weighted Imaging: Systematic Review and Meta-analysis - ScienceDirect." Accessed: Apr. 19, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1878875020317502

[10] N. Galldiks, "Diagnosing pseudoprogression in glioblastoma: A challenging clinical issue," Neuro-Oncol. Pract., vol. 11, no. 1, pp. 1–2, Feb. 2024, doi: 10.1093/nop/npad078.

[11] J. A. Patricoski-Chavez, S. Nagpal, R. Singh, J. L. Warner, and E. D. Gamsiz Uzun, "A deep learning model to predict glioma recurrence using integrated genomic and clinical data," Commun. Med., vol. 5, no. 1, p. 359, Aug. 2025, doi: 10.1038/s43856-025-01083-3.

[12] D. Lteif, D. Appapogu, S. A. Bargal, B. A. Plummer, and V. B. Kolachalama, "Anatomy-Guided, Modality-Agnostic Segmentation of Neuroimaging Abnormalities," Hum. Brain Mapp., vol. 46, no. 14, p. e70329, Sep. 2025, doi: 10.1002/hbm.70329.

[13] D. B. Saridede, R. N. Kaçmaz, and S. Cengiz, "Enhanced Tumor Classification via Radiomics and Deep Feature Generation using Neural Network : Integrating Radiomics with CNN-Based Deep Features," in 2025 Medical Technologies Congress (TIPTEKNO), Oct. 2025, pp. 1–4. doi: 10.1109/TIPTEKNO68206.2025.11270196.

[14] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. Roth, and D. Xu, "Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images," arXiv.org. Accessed: Apr. 24, 2026. [Online]. Available: https://arxiv.org/abs/2201.01266v1

[15] N. Neverova, C. Wolf, G. Taylor, and F. Nebout, "ModDrop: Adaptive Multi-Modal Gesture Recognition," IEEE Trans. Pattern Anal. Mach. Intell., vol. 38, no. 8, pp. 1692-1706, Aug. 2016, doi: 10.1109/TPAMI.2015.2461544.

[16] U.S. Food and Drug Administration, "Transparency for Machine Learning-Enabled Medical Devices: Guiding Principles," FDA, Silver Spring, MD, USA, Jun. 2024. [Online]. Available: https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-software-medical-device